Introduction to Programming with Scientific Applications

- Missing handins, deadline 15 May 2026
- Final project, deadline 31 May 2026
- Course evaluation
- Exam, 20 June 2026
- AOB

Exam

- **5 hours, written exam, with aids, including PC but without internet**
- **Communication with others about the exam is not permitted during the exam**
- **WISEflow Device Monitor must be enabled**
- **AI assistants are not allowed**
- Reexam in August
- Grade is an *overall assessment* of the implementation project and the exam
  - The result of the final exam must meet the minimum requirements for acceptance to be able to pass the course
  - The final exam contributes 80% to the final grade
- wiseflow.au.dk
  - Download .zip + add missing code + upload new .zip
- Questions? – post them on Brightspace

Content of .zip file

A.py - C:\Users\au121\Desktop\ipsa22\ipsa22exam\A.py (3.11.3)

File Edit Format Run Options Window Help

```python
'''
    INTERVAL SUM

    Your task is to write a function interval_sum(i, j), that returns
    the sum i + (i + 1) + ... + j. Eg. for i = 10 and j = 13 the sum
    returned should be 10 + 11 + 12 + 13 = 46.

    Input:  Two lines, containing integers i and j, respectively.
            It is guaranteed that 1 <= i <= j <= 100.

    Output: The sum i + (i + 1) + ... + j.

    Example:

      Input:  10
              13

      Output: 46

    Note: The below code already handles the input and ouput.
'''


def interval_sum(i, j):
    # insert code
    pass


i = int(input())
j = int(input())
print(interval_sum(i, j))
```

Ln: 1 Col: 0

Exam questions

- tests
  - A
    - A.in1
    - A.in2
    - A.in3
    - A.in4
    - A.in5
    - A.out1
    - A.out2
    - A.out3
    - A.out4
    - A.out5
  - B
  - C
  - D
  - E
  - F
  - G
  - H
  - I
  - J
  - K
  - L
- __README__.txt
- A.py
- B.py
- C.py
- D.py
- E.py
- F.py
- G.py
- H.py
- I.py
- J.py
- K.py
- L.py
- run_tests.py

A.in1

Fil Rediger Vis

```
10
13
```

Ln 1, Col 1 100% Windows (CRLF) UTF-8

test input

A.out1

Fil Rediger Vis

```
46
```

Ln 1, Col 1 100% Windows (CRLF) UTF-8

test output

Command Prompt

```
(c) Microsoft Corporation. All rights reserved.

C:\Users\au121>cd Desktop\ipsa22\ipsa22exam

C:\Users\au121\Desktop\ipsa22\ipsa22exam>python run_tests.py A --verbose=-1

Tests passed:

  A       0/5
  -----------
  Total   0/5
  ===========

C:\Users\au121\Desktop\ipsa22\ipsa22exam>python run_tests.py A --verbose=3 --abort
A.py tests/A\A.in1 [failed]
  Input
  > 10
  > 13
  Correct output
  > 46
  Received output
*> None

C:\Users\au121\Desktop\ipsa22\ipsa22exam>
```

run_tests.py arguments

- A B C …
  exercises to evaluate
  default are all
- --abort
  stop on first error
- --verbose=*value*
  amount of output

* = line with wrong output

Using VS Code

F11 toggles full screen

Place terminal right

Right click menu > Panel position > Right

ctrl-k ctrl-o
Open folder

Click to toggle file explorer

Disable AI assistants (GitHub Copilot)

ipsa25exam

EXPLORER

IPSA25EXAM

- tests
- __README__.txt
- A.py
- B.py
- C.py
- D.py
- E.py
- F.py 2
- G.py
- H.py
- I.py
- J.py
- K.py
- L.py
- run_tests.log
- run_tests.py

OUTLINE

TIMELINE

F.py 2

F.py > ...

```python
'''
    PROGRESSION

    Your task is to create a function progression(L), that takes a list L of one
    of the following three formats:

        [first, ..., end]
        [first, second, ..., end]
        [first, second, third, ..., end]

    where first, second, and third are integers, and ... is the Ellipsis object.

    The function progression should return a list of integers, such that in the
    first case it just returns all integers from first to end in increasing
    order, including end (it is guaranteed that first < end). E.g.,

        progression([5, ..., 15]) = [5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15]

    Note: The below code already reads the input and prints the result of
          calling the function progression.
'''


def progression(L):
    # insert code
    pass

    return list(range(L[0], L[2] + 1))  # only works for 1st test...


print(progression(eval(input())))
```

Split tab

hold “alt” to split horizontally

PROBLEMS 5 TERMINAL powershell

```
PS C:\Users\au121\OneDrive - Aarhus universitet\Skrivebord\ipsa25\ipsa
25exam> python run_tests.py --verbose=-1

Tests passed:

  A.py    0/5
  B.py    0/5
  C.py    0/5
  D.py    0/5
  E.py    0/5
  F.py    1/5
  G.py    0/5
  H.py    0/5
  I.py    0/5
  J.py    0/5
  K.py    0/5
  L.py    0/5
  ------------
  Total   1/60
  ============
PS C:\Users\au121\OneDrive - Aarhus universitet\Skrivebord\ipsa25\ipsa
25exam> python run_tests.py --verbose=3 --abort F
F.py tests/F\F.in1 [ok]
  Input
  > [5, ..., 15]
  Correct output
  > [5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15]
F.py tests/F\F.in2 [failed]
  Input
  > [4, 7, ..., 31]
  Correct output
  > [4, 7, 10, 13, 16, 19, 22, 25, 28, 31]
  Received output
  > (none)
  Error
  > ~~~~~~~~~~~^^^^^^^^^^^^^^
  > File "C:\Users\au121\OneDrive - Aarhus universitet\Skrivebord\ipsa
25\ipsa25exam\F.py", line 61, in progression
  > return list(range(L[0], L[2] + 1)) # only works for 1st test...
  > ~~~~~^~~
  > TypeError: unsupported operand type(s) for +: 'ellipsis' and 'int'
PS C:\Users\au121\OneDrive - Aarhus universitet\Skrivebord\ipsa25\ipsa
25exam>
```

Ctrl-j
Toggles terminal

Ctrl-l
Clears terminal

0 2 3 Ln 45, Col 1 Spaces: 4 UTF-8 CRLF {} Python Python 3.14.3

Evaluation of code

Don’t expect partial scores for this solution

```
def interval_sum(i, j):
    if i == 10 and j == 13:
        return 46
```

- Each problem will be assigned a **weight**
- There will be problems of **varying difficulty**
- Code will be evaluated on **known test cases** and **unknown test cases**
- In general, **automatic scoring**, in some exceptional cases manual

- Put a **comment if you copied code from exercises/slides/documentation** to avoid plagiarism

AOB ?